

CPU Scheduling Algorithms

Performance Analysis Report

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Input Processes

Process ID	Arrival Time	Burst Time	Priority
P1	0	8	3
P2	1	7	1
P3	2	9	4
P4	3	5	2
P5	4	2	3

Time Quantum (Round Robin): 2 time units

FCFS

PID	AT	BT	CT	TAT	WT	RT
P1	0	8	8	8	0	0
P2	1	7	15	14	7	7
P3	2	9	24	22	13	13
P4	3	5	29	26	21	21
P5	4	2	31	27	25	25

P1(0-8) | P2(8-15) | P3(15-24) | P4(24-29) | P5(29-31)

Metric	Value
Average Waiting Time	13.20 units
Average Turnaround Time	19.40 units
Average Response Time	13.20 units
CPU Utilization	100.00%
Throughput	0.1613 processes/unit
Total Execution Time	31 units

SJF (Non-Preemptive)

PID	AT	BT	CT	TAT	WT	RT
P1	0	8	8	8	0	0
P2	1	7	22	21	14	14
P3	2	9	31	29	20	20
P4	3	5	15	12	7	7
P5	4	2	10	6	4	4

P1(0-8) | P5(8-10) | P4(10-15) | P2(15-22) | P3(22-31)

Metric	Value
Average Waiting Time	9.00 units
Average Turnaround Time	15.20 units
Average Response Time	9.00 units
CPU Utilization	100.00%
Throughput	0.1613 processes/unit
Total Execution Time	31 units

Round Robin (Q=2)

PID	AT	BT	CT	TAT	WT	RT
P1	0	8	20	20	12	0
P2	1	7	28	27	20	3
P3	2	9	31	29	20	4
P4	3	5	27	24	19	7
P5	4	2	14	10	8	8

P1(0-4) | P2(4-6) | P3(6-8) | P1(8-10) | P4(10-12) | P5(12-14) | P1(14-16) |
P2(16-18) | P3(18-20) | P4(20-22) | P2(22-24) | P3(24-26) | P4(26-27) | P2(27-28) |
P3(28-31)

Metric	Value
Average Waiting Time	15.80 units
Average Turnaround Time	22.00 units
Average Response Time	4.40 units
CPU Utilization	100.00%
Throughput	0.1613 processes/unit
Total Execution Time	31 units

Priority (Non-Preemptive)

PID	AT	BT	CT	TAT	WT	RT
P1	0	8	8	8	0	0
P2	1	7	15	14	7	7
P3	2	9	31	29	20	20
P4	3	5	20	17	12	12
P5	4	2	22	18	16	16

P1(0-8) | P2(8-15) | P4(15-20) | P5(20-22) | P3(22-31)

Metric	Value
Average Waiting Time	11.00 units
Average Turnaround Time	17.20 units
Average Response Time	11.00 units
CPU Utilization	100.00%

Throughput	0.1613 processes/unit
Total Execution Time	31 units

Algorithms Comparison Summary

Algorithm	Avg WT	Avg TAT	Avg RT	CPU%
FCFS	13.20	19.40	13.20	100.0%
SJF (Non-Preemptive)	9.00	15.20	9.00	100.0%
Round Robin (Q=2)	15.80	22.00	4.40	100.0%
Priority (Non-Preemptive)	11.00	17.20	11.00	100.0%

Analysis

Best Average Waiting Time: SJF (Non-Preemptive) (9.00 units)

Best Average Turnaround Time: SJF (Non-Preemptive) (15.20 units)

Best Average Response Time: Round Robin (Q=2) (4.40 units)

Recommendation: The choice of algorithm depends on system requirements. SJF generally minimizes waiting time but can cause starvation. Round Robin provides fair CPU allocation and is ideal for time-sharing systems. FCFS is simple but can lead to the convoy effect. Priority scheduling is useful when tasks have different importance levels.